

Calibration results

Normalized Residuals

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Reprojection error (cam0):    mean 1.0412102665331984, median 0.9076394296732813, std: 0.6775853101247179
Reprojection error (cam1):    mean 0.9938837843745534, median 0.843183993368082, std: 0.6793929107413212
Gyroscope error (imu0):      mean 1.763013815094, median 1.5372008477792884, std: 1.0234618415487797
Accelerometer error (imu0):  mean 0.4645965859245904, median 0.4004885384928988, std: 0.36347265556257635
```

Residuals

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Reprojection error (cam0) [px]:    mean 1.0412102665331984, median 0.9076394296732813, std: 0.6775853101247179
Reprojection error (cam1) [px]:    mean 0.9938837843745534, median 0.843183993368082, std: 0.6793929107413212
Gyroscope error (imu0) [rad/s]:    mean 0.017630138150940004, median 0.015372008477792885, std: 0.010234618415487798
Accelerometer error (imu0) [m/s^2]: mean 0.18583863436983616, median 0.1601954153971595, std: 0.14538906222503054
```

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[ -0.00066472  0.02663794  0.99964493  0.01171121]
 [ -0.00163857  0.99964378 -0.02663899  0.16855436]
 [ -0.99999844 -0.0016557  -0.00062084 -0.11534674]
 [  0.          0.          0.          1.          ]]
```

T_ic: (cam0 to imu0):

```
[[ -0.00066472 -0.00163857 -0.99999844 -0.11506258]
 [  0.02663794  0.99964378 -0.0016557  -0.16899726]
 [  0.99964493 -0.02663899 -0.00062084 -0.00728855]
 [  0.          0.          0.          1.          ]]
```

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.0002900309601826913

Transformation (cam1):

T_ci: (imu0 to cam1):

```
[[  0.01716141  0.02216055  0.99960712 -0.03696207]
 [ -0.00526806  0.99974248 -0.02207311  0.1683525 ]
 [ -0.99983885 -0.00488719  0.01727373 -0.11662901]
 [  0.          0.          0.          1.          ]]
```

```
T_ic: (cam1 to imu0):  
[[ 0.01716141 -0.00526806 -0.99983885 -0.115089 ]  
 [ 0.02216055 0.99974248 -0.00488719 -0.16806004]  
 [ 0.99960712 -0.02207311 0.01727373 0.04267823]  
 [ 0. 0. 0. 1. ]]
```

```
timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)  
-0.00029670492506792093
```

Baselines:

```
Baseline (cam0 to cam1):  
[[ 0.99983109 -0.00450399 -0.01781867 -0.04996747]  
 [ 0.0045693 0.99998298 0.00362649 0.0001658 ]  
 [ 0.01780203 -0.00370729 0.99983466 -0.00088495]  
 [ 0. 0. 0. 1. ]]
```

```
baseline norm: 0.049975578152190045 [m]
```

```
Gravity vector in target coords: [m/s^2]  
[ 0.26672908 -9.80291914 -0.00740525]
```

Calibration configuration

=====

cam0

```
Camera model: omni  
Focal length: [2044.6753301041977, 2065.5927559749084]  
Principal point: [616.3310327581959, 404.1255934969876]  
Omni xi: 1.5936757817618359  
Distortion model: radtan  
Distortion coefficients: [2.290964717370836, -1.8671244246514953, -0.010010836103043635, 0.002831387673779597]  
Type: aprilgrid  
Tags:  
  Rows: 6  
  Cols: 6  
  Size: 0.02 [m]  
  Spacing 0.006 [m]
```

cam1

```
-----
Camera model: omni
Focal length: [2037.2418743359228, 2054.1495623160185]
Principal point: [634.1340582275024, 379.92908128448573]
Omni xi: 1.569995877298825
Distortion model: radtan
Distortion coefficients: [2.167702619581351, -0.9560356055374504, -0.01088770643867036, 0.012396986450845738]
Type: aprilgrid
Tags:
  Rows: 6
  Cols: 6
  Size: 0.02 [m]
  Spacing 0.006 [m]
```

IMU configuration

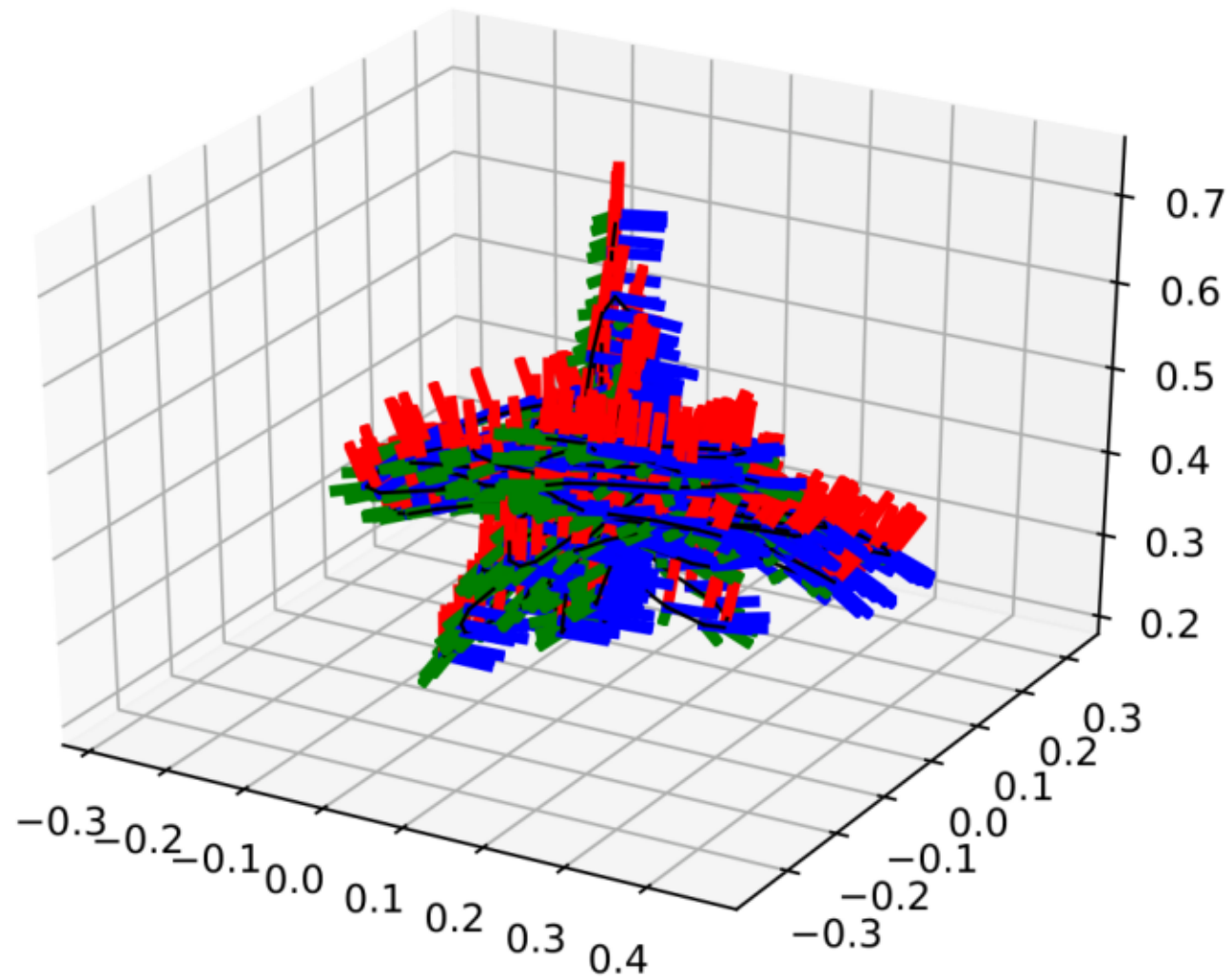
=====

IMU0:

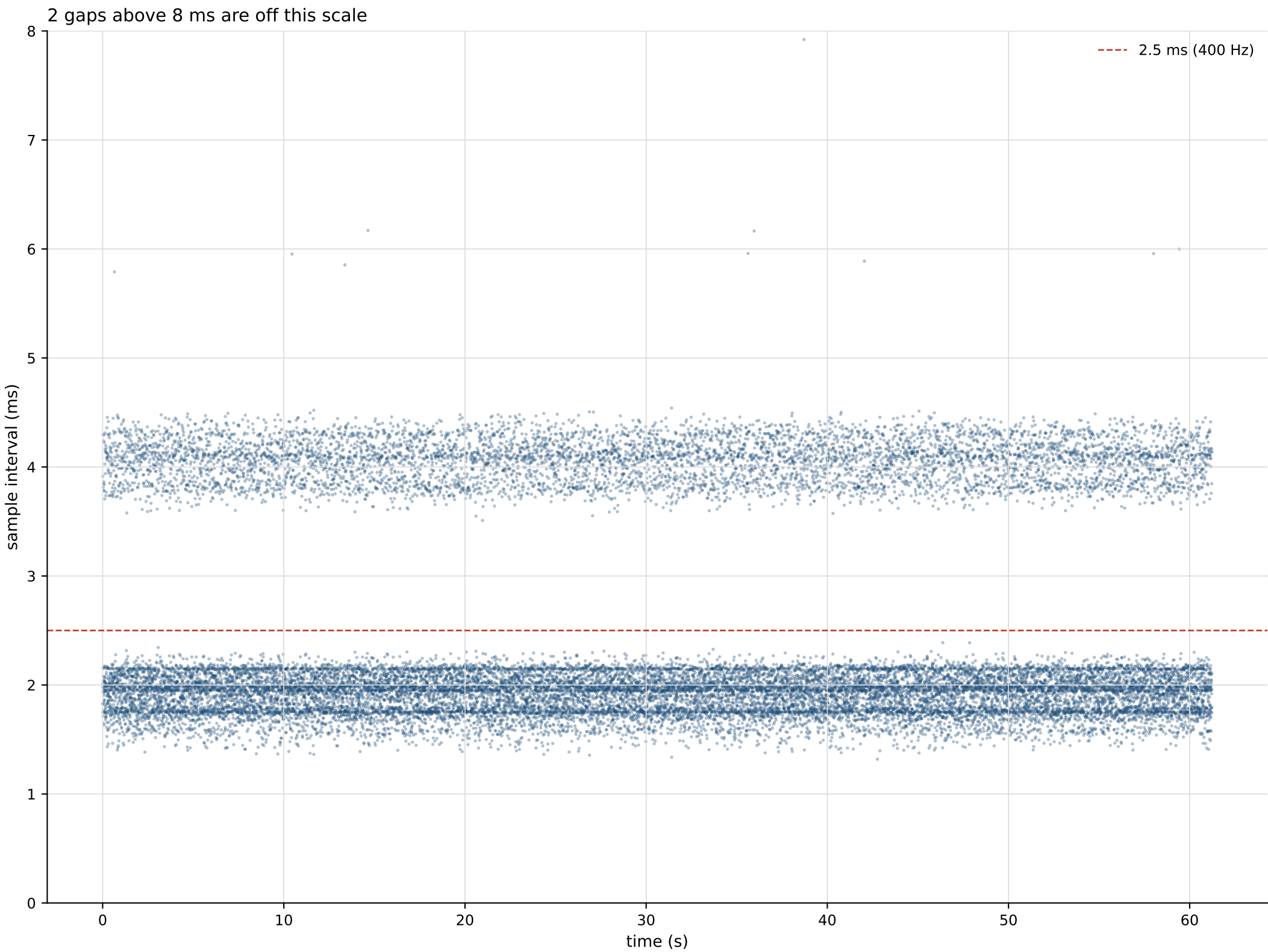
```
-----
Model: calibrated
Update rate: 400
Accelerometer:
  Noise density: 0.02
  Noise density (discrete): 0.39999999999999997
  Random walk: 0.001
Gyroscope:
  Noise density: 0.0005
  Noise density (discrete): 0.01
  Random walk: 0.0001
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]
```

IMU estimated poses

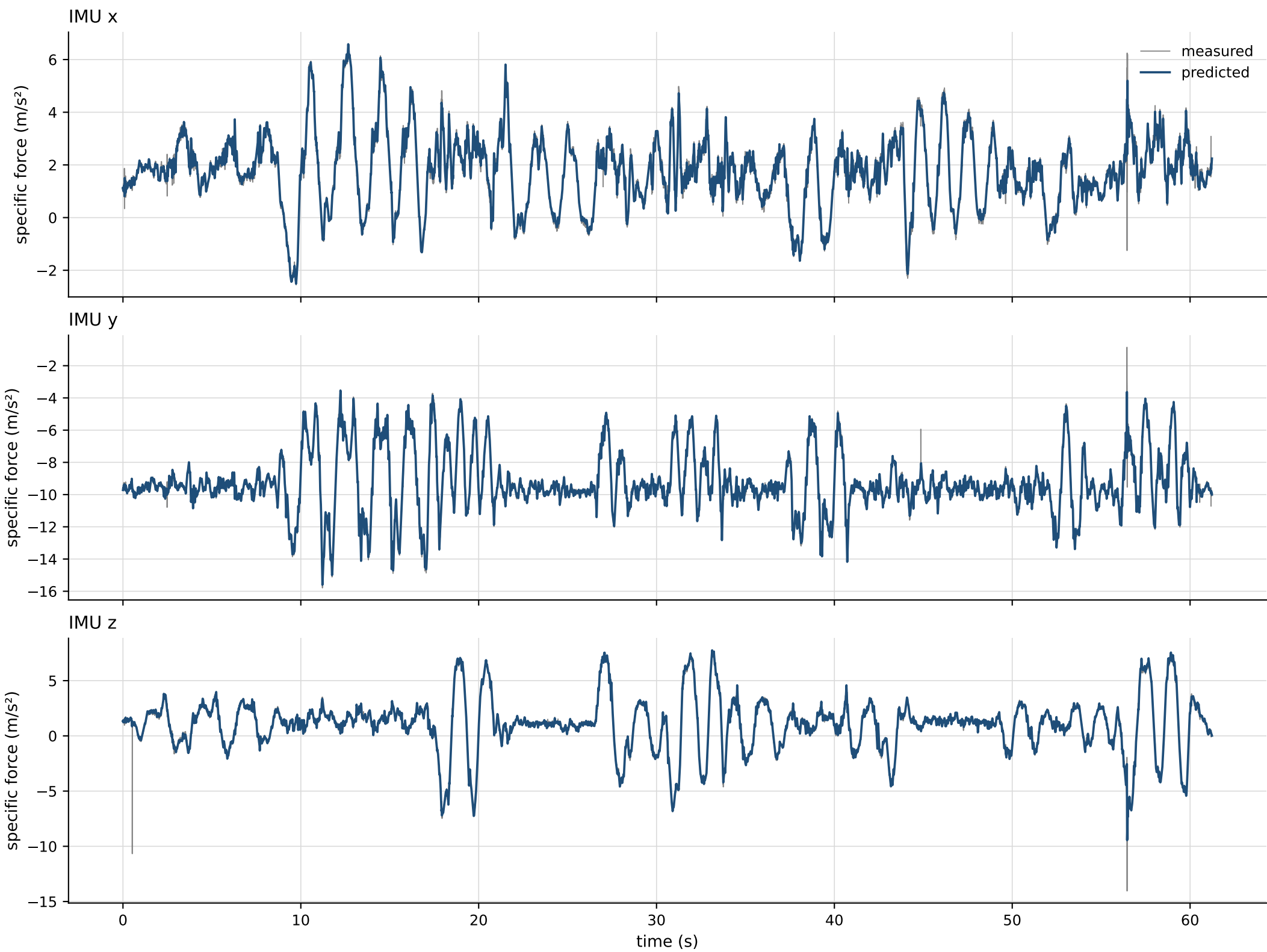
imu0: estimated poses



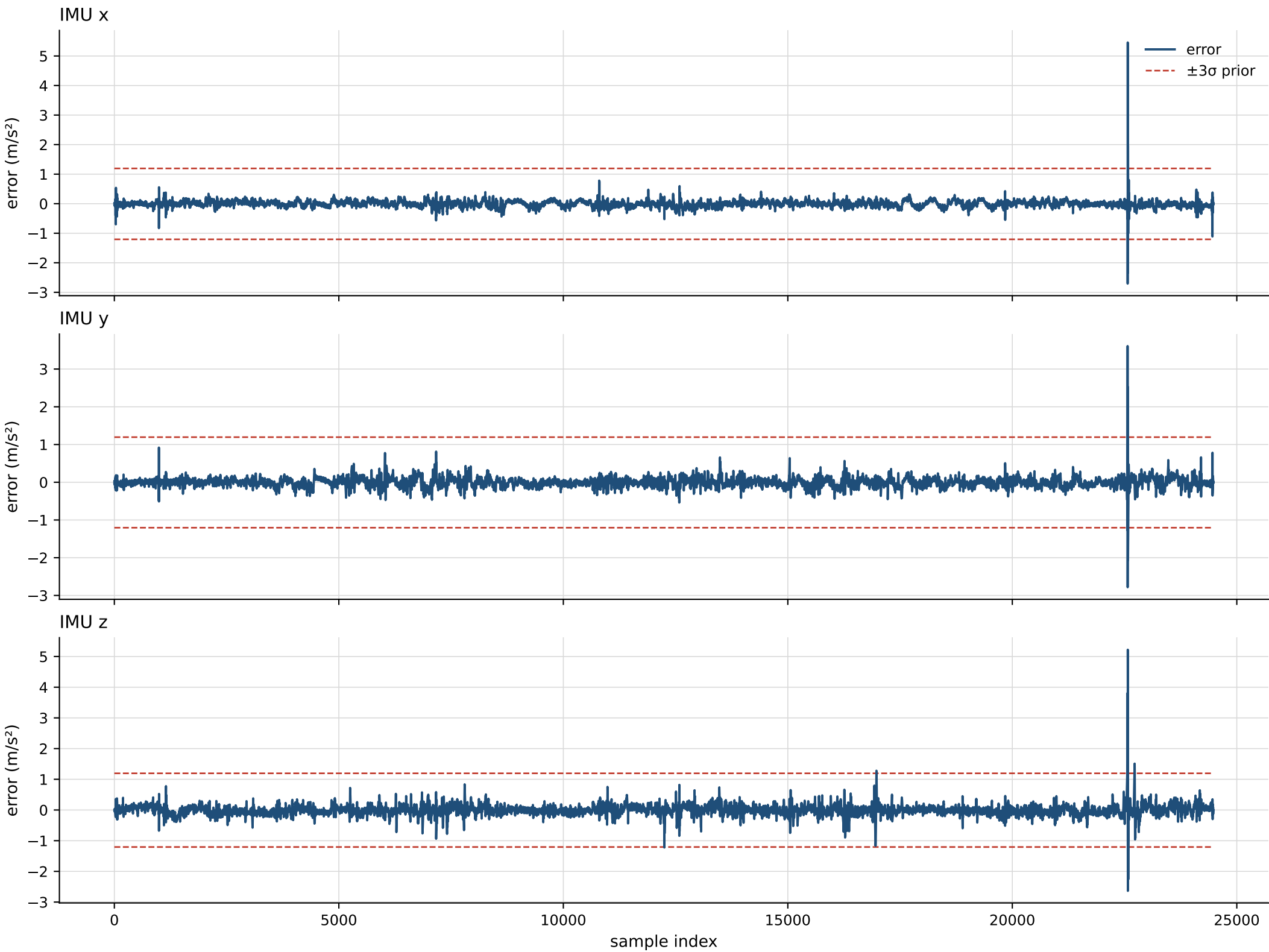
IMU sample interval



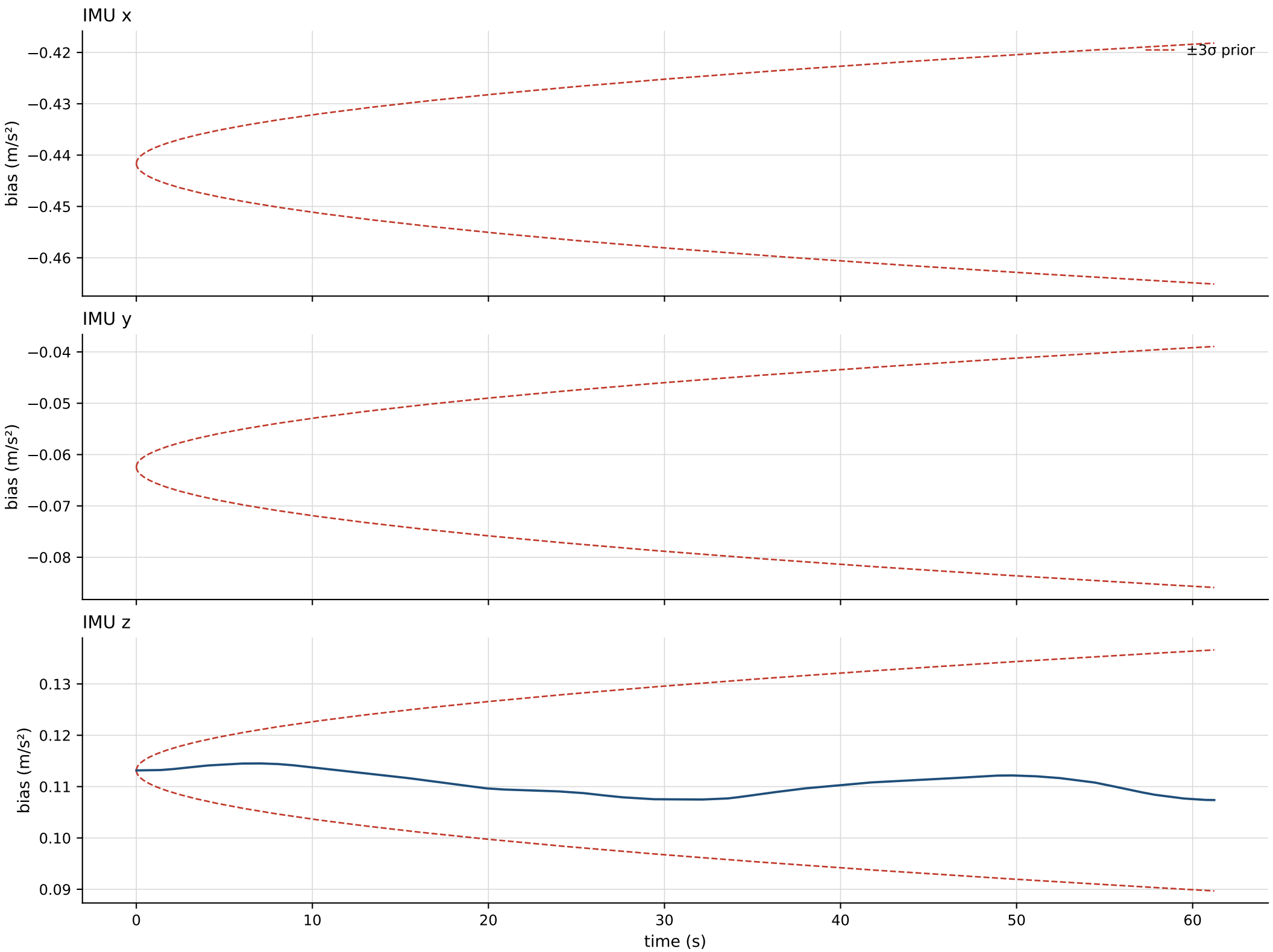
Specific force: predicted motion vs measured accelerometer



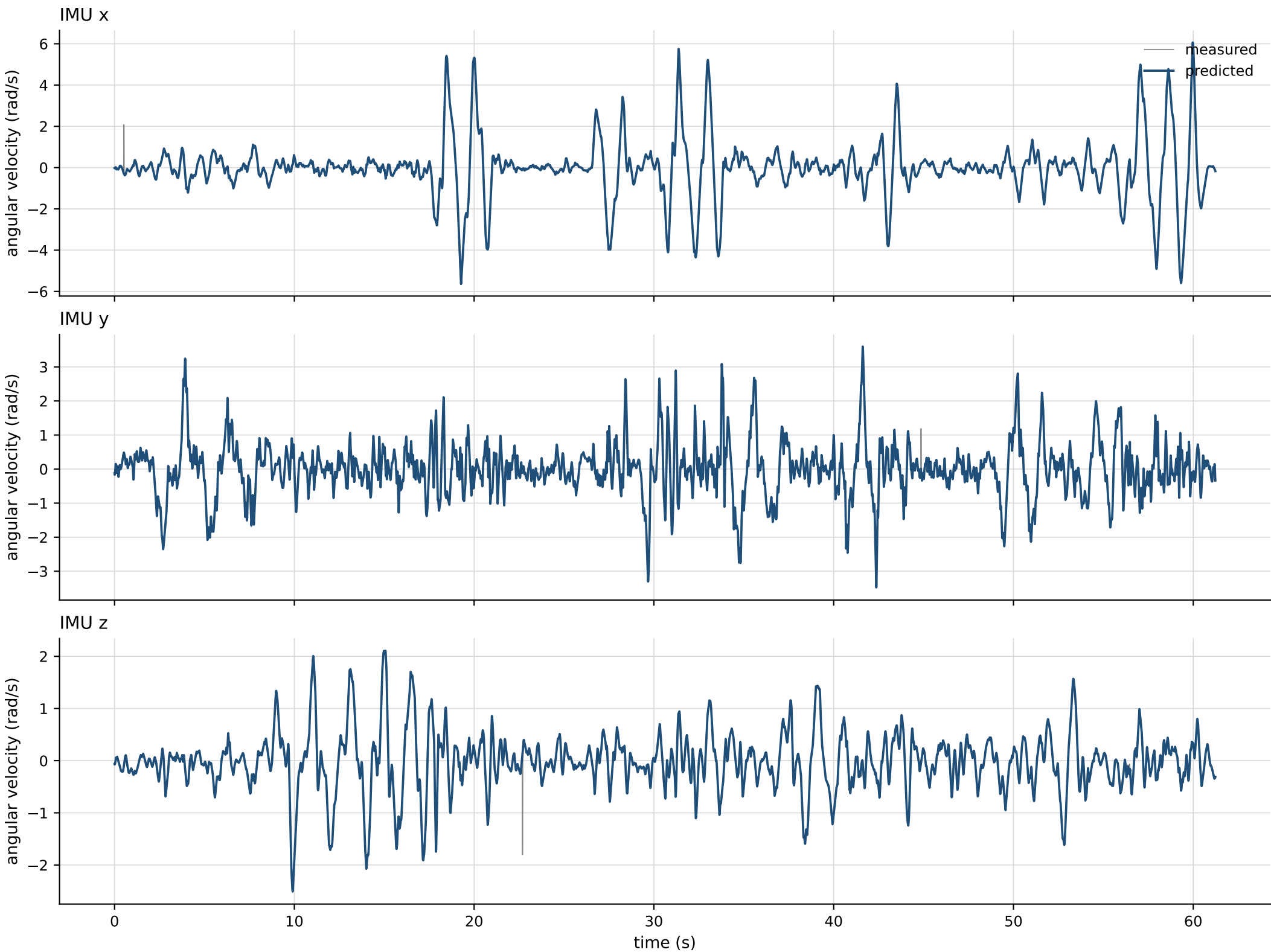
Accelerometer error



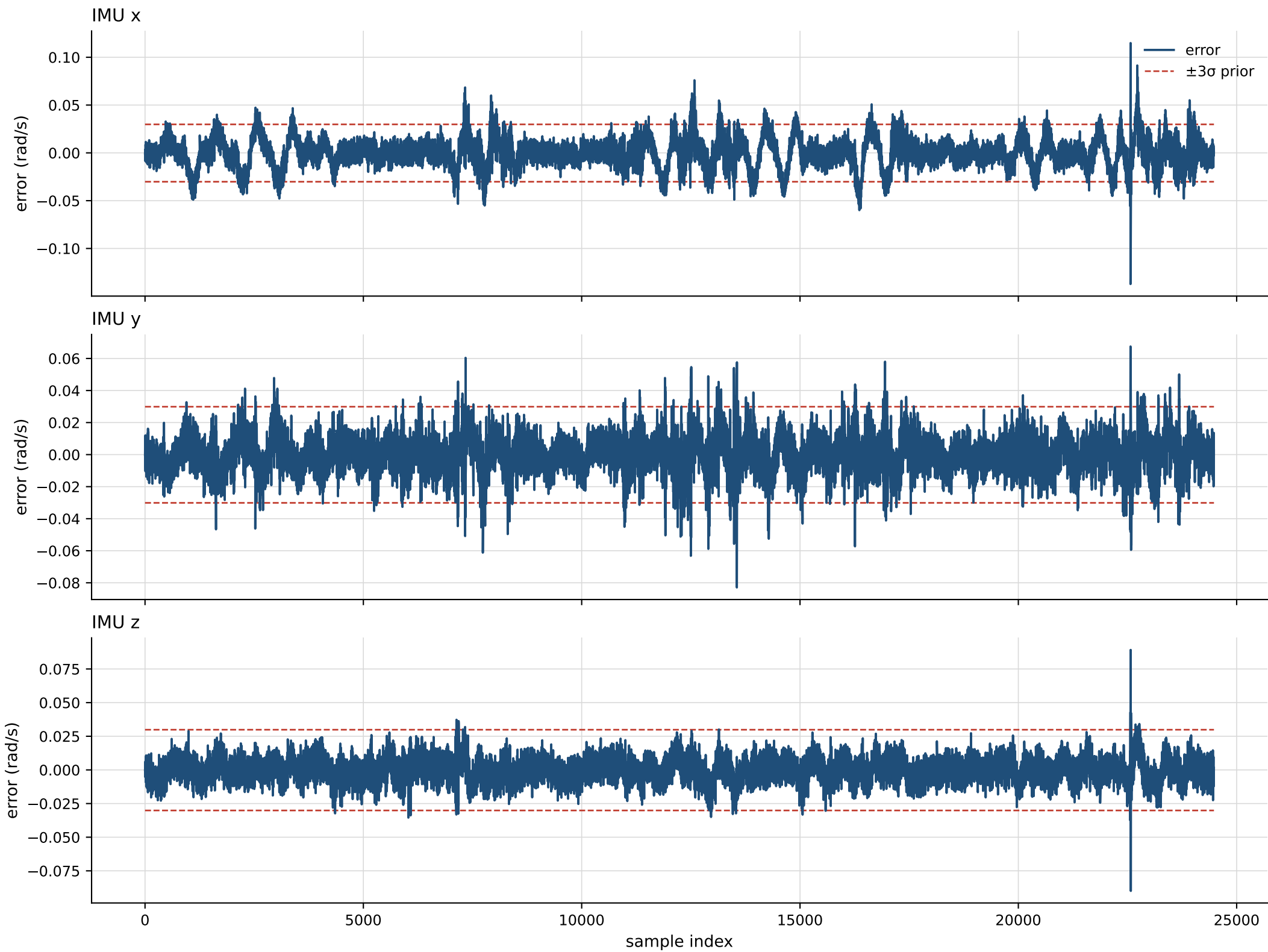
Estimated accelerometer bias



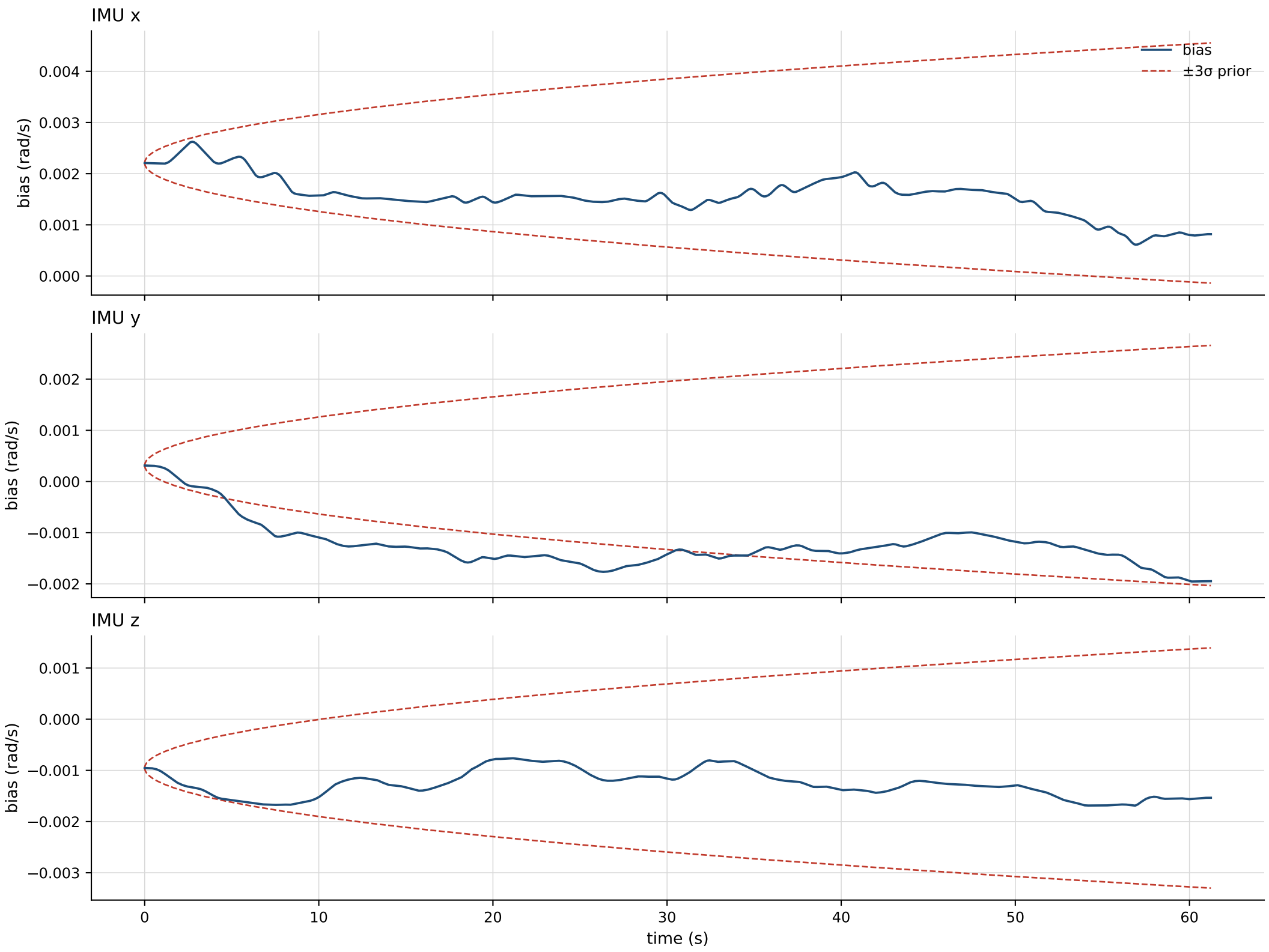
Angular velocity: predicted motion vs measured gyroscope



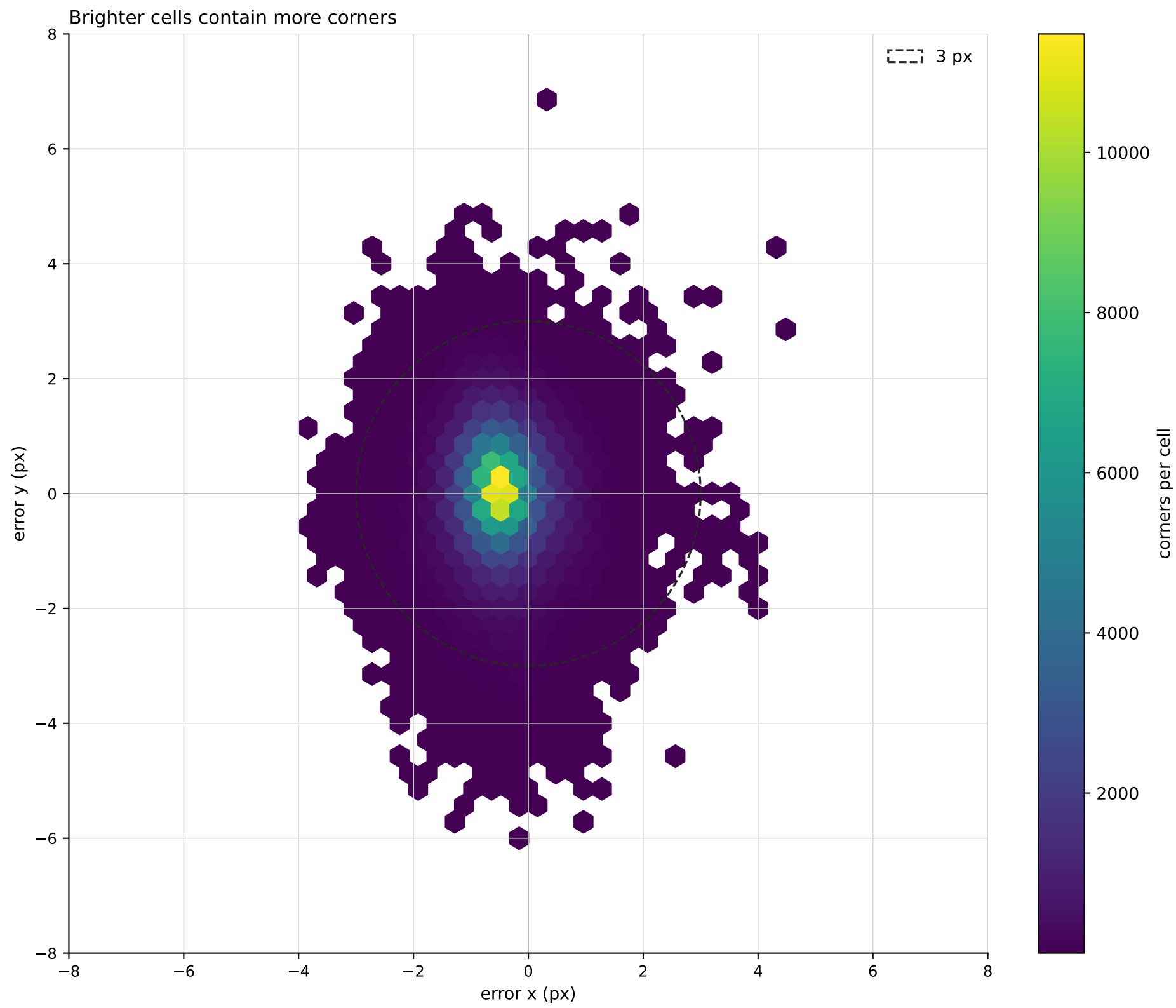
Gyroscope error



Estimated gyroscope bias



Left camera: reprojection error



Right camera: reprojection error

